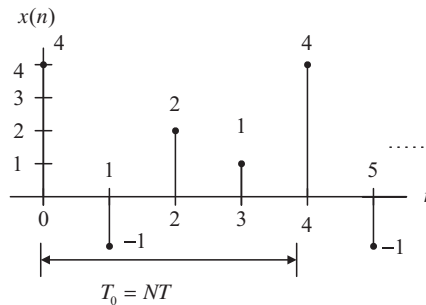

4.6 SUMMARY

1. The Fourier series coefficients for a periodic digital signal can be used to develop the DFT.
 2. The DFT transforms a time sequence to the complex DFT coefficients, while the inverse DFT transforms DFT coefficients back to the time sequence.
 3. The *frequency bin number* is the same as the frequency index. *Frequency resolution* is the frequency spacing between two consecutive frequency indices (two consecutive spectrum components).
 4. The DFT coefficients for a given digital signal are applied to compute the amplitude spectrum, power spectrum, or phase spectrum.
 5. The spectrum calculated from all the DFT coefficients represents the signal frequency range from 0 Hz to the sampling rate. The spectrum beyond the folding frequency is equivalent to the negative-indexed spectrum from the negative folding frequency to 0 Hz. This two-sided spectrum can be converted into a single-sided spectrum by doubling alternation-current (AC) components from 0 Hz to the folding frequency and retaining the DC component as is.
 6. To reduce the burden of computing DFT coefficients, the FFT algorithm is used, which requires the data length to be a power of 2. Sometimes zero padding is employed to make up the data length. The zero padding actually interpolates the spectrum and does not carry any new information about the signal; even the calculated frequency resolution is smaller due to the zero-padded longer length.
 7. Applying a window function to the data sequence before DFT reduces the spectral leakage due to abrupt truncation of the data sequence when performing spectral calculation for a short sequence.
 8. Two radix-2 FFT algorithms—decimation-in-frequency and decimation-in-time—are developed via graphical illustrations.
-

4.7 PROBLEMS

- 4.1. Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 1$, $x(1) = 1$, $x(2) = -1$, and $x(3) = 0$, compute its DFT $X(k)$.
- 4.2. Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 4$, $x(1) = 3$, $x(2) = 2$, and $x(3) = 1$, evaluate its DFT $X(k)$.
- 4.3. Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 0.2$, $x(1) = 0.2$, $x(2) = -0.2$, and $x(3) = 0$, compute its DFT $X(k)$.
- 4.4. Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 0.8$, $x(1) = 0.6$, $x(2) = 0.4$, and $x(3) = 0.2$, evaluate its DFT $X(k)$.
- 4.5. Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ obtained in Problem 4.2, evaluate its inverse DFT $x(n)$.
- 4.6. Given a sequence $x(n)$, where $x(0) = 4$, $x(1) = 3$, $x(2) = 2$, and $x(3) = 1$ with two additional zero-padded data points $x(4) = 0$ and $x(5) = 0$, evaluate its DFT $X(k)$.
- 4.7. Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ obtained in Problem 4.4, evaluate its inverse DFT $x(n)$.

- 4.8.** Given a sequence $x(n)$, where $x(0) = 0.8$, $x(1) = 0.6$, $x(2) = 0.4$, and $x(3) = 0.2$ with two additional zero-padded data points $x(4) = 0$ and $x(5) = 0$, evaluate its DFT $X(k)$.
- 4.9.** Using the DFT sequence $X(k)$ for $0 \leq k \leq 5$ computed in Problem 4.6, evaluate the inverse DFT for $x(0)$ and $x(4)$.
- 4.10.** Consider a digital sequence sampled at the rate of 20,000 Hz. If we use the 8,000-point DFT to compute the spectrum, determine
- the frequency resolution;
 - the folding frequency in the spectrum.
- 4.11.** Using the DFT sequence $X(k)$ for $0 \leq k \leq 5$ computed in Problem 4.8, evaluate the inverse DFT for $x(0)$ and $x(4)$.
- 4.12.** Consider a digital sequence sampled at the rate of 16,000 Hz. If we use the 4,000-point DFT to compute the spectrum, determine
- the frequency resolution;

**FIGURE 4.46**

Data sequence for Problem 4.14.

- the folding frequency in the spectrum.
- 4.13.** We use the DFT to compute the amplitude spectrum of a sampled data sequence with a sampling rate $f_s = 2,000$ Hz. It requires the frequency resolution to be less than 0.5 Hz. Determine the number of data points used by the FFT algorithm and actual frequency resolution in Hz, assuming that the data samples are available for selecting the number of data points.
- 4.14.** Given the sequence in Figure 4.46 and assuming $f_s = 100$ Hz, compute the amplitude spectrum, phase spectrum, and power spectrum.
- 4.15.** Compute the following window functions for a size of eight:
- Hamming window function;
 - Hanning window function.

- 4.16. Consider the following data sequence of length six:

$$x(0) = 0, x(1) = 1, x(2) = 0, x(3) = -1, x(4) = 0, x(5) = 1$$

Compute the windowed sequence $x_w(n)$ using the

- triangular window function;
- Hamming window function;
- Hanning window function.

- 4.17. Compute the following window functions for a size of 10:

- Hamming window function;
- Hanning window function.

- 4.18. Consider the following data sequence of length six:

$$x(0) = 0, x(1) = 0.2, x(2) = 0, x(3) = -0.2, x(4) = 0, x(5) = 0.2$$

Compute the windowed sequence $x_w(n)$ using the

- triangular window function;
- Hamming window function;
- Hanning window function.

- 4.19. Given the sequence in Figure 4.47 where $f_s = 100$ Hz and $T = 0.01$ sec., compute the amplitude spectrum, phase spectrum, and power spectrum using the

- triangular window;
- Hamming window;
- Hanning window.

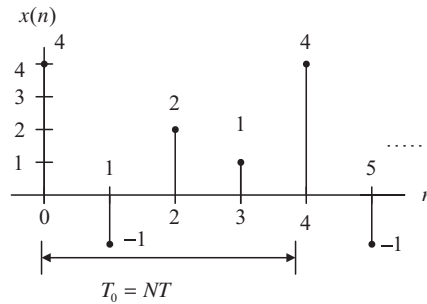


FIGURE 4.47

Data sequence for Problem 4.19.

- 4.20. Given the sinusoid

$$x(n) = 2 \cdot \sin\left(2,000 \cdot 2\pi \cdot \frac{n}{8,000}\right)$$

obtained using a sampling rate of $f_s = 8,000$ Hz, we apply the DFT to compute the amplitude spectrum.

- a. Determine the frequency resolution when the data length is 100 samples. Without using the window function, is there any spectral leakage in the computed spectrum? Explain.
 - b. Determine the frequency resolution when the data length is 73 samples. Without using the window function, is there any spectral leakage in the computed spectrum? Explain.
- 4.21.** Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 4$, $x(1) = 3$, $x(2) = 2$, and $x(3) = 1$, evaluate its DFT $X(k)$ using the decimation-in-frequency FFT method, and determine the number of complex multiplications.
- 4.22.** Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ obtained in Problem 4.21, evaluate its inverse DFT $x(n)$ using the decimation-in-frequency FFT method.
- 4.23.** Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 0.8$, $x(1) = 0.6$, $x(2) = 0.4$, and $x(3) = 0.2$, evaluate its DFT $X(k)$ using the decimation-in-frequency FFT method, and determine the number of complex multiplications.
- 4.24.** Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ obtained in Problem 4.23, evaluate its inverse DFT $x(n)$ using the decimation-in-frequency FFT method.
- 4.25.** Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 4$, $x(1) = 3$, $x(2) = 2$, and $x(3) = 1$, evaluate its DFT $X(k)$ using the decimation-in-time FFT method, and determine the number of complex multiplications.
- 4.26.** Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ computed in Problem 4.25, evaluate its inverse DFT $x(n)$ using the decimation-in-time FFT method.
- 4.27.** Given a sequence $x(n)$ for $0 \leq n \leq 3$, where $x(0) = 0.8$, $x(1) = 0.4$, $x(2) = -0.4$, and $x(3) = -0.2$, evaluate its DFT $X(k)$ using the decimation-in-time FFT method, and determine the number of complex multiplications.
- 4.28.** Given the DFT sequence $X(k)$ for $0 \leq k \leq 3$ computed in Problem 4.27, evaluate its inverse DFT $x(n)$ using the decimation-in-time FFT method.

4.7.1 Computer Problems with MATLAB

Use MATLAB to solve Problems 4.29 and 4.30.

- 4.29.** Consider three sinusoids with the following amplitudes and phases:

$$\begin{aligned}x_1(t) &= 5\cos(2\pi(500)t) \\x_2(t) &= 5\cos(2\pi(1200)t + 0.25\pi) \\x_3(t) &= 5\cos(2\pi(1800)t + 0.5\pi)\end{aligned}$$

- a. Create a MATLAB program to sample each sinusoid and generate a sum of three sinusoids, that is, $x(n) = x_1(n) + x_2(n) + x_3(n)$, using a sampling rate of 8,000 Hz. Plot $x(n)$ over a range of 0.1 seconds.
- b. Use the MATLAB function `fft()` to compute DFT coefficients, and plot and examine the spectrum of the signal $x(n)$.

4.30. Consider the sum of sinusoids in Problem 4.29.

- a. Generate the sum of sinusoids for 240 samples using a sampling rate of 8,000 Hz.
- b. Write a MATLAB program to compute and plot the amplitude spectrum of the signal $x(n)$ with the FFT using each of the following window functions:
 - (1) Rectangular window (no window);
 - (2) Triangular window;
 - (3) Hamming window.
- c. Examine the effect of spectral leakage for each window use in (b).

4.7.2 MATLAB Projects

4.31. Signal spectral analysis:

Given the four practical signals below, compute their one-sided spectra and create their time-domain plots and spectral plots, respectively:

- a. Speech signal (“speech.dat”), sampling rate = 8,000 Hz. From the spectral plot, identify the first 5 formants.
- b. ECG signal (“ecg.dat”), sampling rate = 500 Hz. From the spectral plot, identify the 60 Hz-interference component.
- c. Seismic data (“seismic.dat”), sampling rate = 15 Hz. From the spectral plot, determine the dominant frequency component.
- d. Vibration signal of the acceleration response from a simple supported beam (“vbrdata.dat”), sampling rate = 1,000 Hz. From the spectral plot, determine four dominant frequencies (modes).

4.32. Vibration signature analysis:

The acceleration signals measured from a gearbox can be used to monitor the condition of the gears inside the gearbox. The early diagnosis of any gear issues can prevent the future catastrophic failure of the system. Assume the following measurements and specifications (courtesy of SpectraQuest, Inc.):

- a. The input shaft has a speed of 1,000 RPM and meshing frequency is approximately 300 Hz.
- b. Data specifications:

Sampling rate = 12.8 kHz

v0.dat: healthy condition

v1.dat: damage severity level 1 (lightly chipped gear)

v2.dat: damage severity level 2 (moderately chipped gear)

v3.dat: damage severity level 3 (chipped gear)

v4.dat: damage severity level 4 (heavily chipped gear)

v5.dat: damage severity level 5 (missing tooth)

Investigate the spectrum for each measurement and identify sidebands. For each measurement, determine the ratio of the largest sideband amplitude over the amplitude of meshing frequency. Investigate the relation between the computed ratio values and the damage severity.